

BCA (Semester-II) Examination, 2022

(Session 2021-23)

DATASTRUCTURE THROUGH "C"

[Paper Code : BC-203]

Time : Three Hours

[Maximum Marks : 80]

Note: Candidates are required to give their answers in their own words as far as practicable. The questions are of equal value. Answer any five questions.

1. What is an Array? Explain the types of Array. Write a C program to sort $m \times n$ matrix in ascending order where 'm' refers to rows and 'n' refers to columns.
2. What is Recursive function? Distinguish between normal function and recursion with suitable example.
3. Explain data structure. Discuss the types of data structure with neat diagram.
4. Write a C program to perform the following operation using single linked list:
 - (a) Create a node
 - (b) Insert a node
 - (c) Display a node
 - (d) Search a specific node
 - (e) Delete a node
 - (f) Exit
5. Write a C program to implement stack operation using structure consist of an array of items.
6. What is Queue? Discuss the types of queue. Explain the delete and insert operation in a queue with diagram.
7. What is Tree? Discuss binary tree traversal recursive algorithm in detail.
8. Explain searching and sorting. Discuss the types of searching with suitable examples.
9. What is Pointer? Write a C program to enter n number in an array. Print maximum and minimum number in the given list of array using pointer.
10. Write notes on any two from the following:
 - (a) Application of data structure
 - (b) Double linked list
 - (c) Binary tree
 - (d) Selection sorting